



SRI SANKARAN

Ph.D. Scholar, Department of Electrical Engineering, IIT Bombay

Experimental physicist with experience in embedded systems, intelligent sensing, and automated instrumentation. Skilled in developing real-time sensing systems integrating embedded hardware, signal acquisition, sensor fusion, and experimental automation.

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EXAMINATION/DEGREE	INSTITUTE	CGPA/%	YEAR
Ph.D. (Electrical Engineering)	Indian Institute of Technology Bombay (IITB), Mumbai	-	July 2026 - Present
Integrated M.Sc. (Physics)	Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat	8.07 CGPA	2025
Higher Secondary (Class XII – Computer Science)	Kuriakose Elias English Medium School, Mannanam (DHSE Kerala)	91.33 %	2020
Secondary School Examination (Class X)	Rajagiri Higher Secondary School, Kalamassery (Kerala State Board – SSLC)	98.79 %	2018

RESEARCH INTERESTS

- MEMS/NEMS & Sensor Instrumentation
- Scientific Instrumentation & Automated Measurement Systems
- Sensor Fusion & Signal Processing
- Real-Time Measurement & Data Acquisition
- Embedded Systems & Edge Computing
- Wearable & Distributed Sensing
- Cyber-Physical Systems (CPS)

RESEARCH EXPERIENCES

- IoT-BASED VENTILATION OF UNIVERSITY CLASSROOMS FOR INDOOR AIR QUALITY & AIRBORNE INFECTION CONTROL – OPTIMIZATION FOR HEALTH, ENERGY & COMFORT**

Technical Contributor – In collaboration with research activities led by Prof. Anupama Kowli | Electrical Engineering Dept, IIT Bombay

Sep 2025 – Dec 2025

- Contributed to an ongoing IoT-based smart ventilation system for university classrooms, with a focus on instrumentation, sensor evaluation, and system-level understanding of health, energy, and thermal comfort parameters.
- Worked with industry-grade environmental sensors for temperature, humidity, CO₂, and particulate matter (PM) monitoring.
- Owled **firmware-level integration and telemetry handling** for selected sensor nodes, implementing robust sensor communication workflows including OTA (over-the-air) firmware updates and Telnet-based remote diagnostics for **in-field updates and live debugging**.
- Implemented sensor communication workflows, real-time data transmission to a central server, and **web-based dashboards** for **live monitoring** and historical **data logging**.
- Provided technical inputs for system updates and data-driven discussions on ventilation performance and optimization.
- Supported **server-related tasks**, including basic setup, maintenance assistance, and exposure to **database handling and visualization** pipelines.
- Contributed to **professional project execution** workflows, including documentation standards, review cycles, and deployment workflows in institutional research-industry collaborations.

- UNDERGRADUATE RESEARCH ASSISTANT**

Optoelectronic Materials and Devices (OMD) Lab | Prof. Vipul Kheraj | SVNIT Surat

June 2022 – May 2025

- Worked extensively on fabrication & testing of **Lead-based & Lead-free (Bi & Sb) perovskite solar cells (PSC)**.
- Hands-on experience** with annealing, spin coating, slot-die coating, UV-vis spectroscopy, PL spectroscopy, Photodetactivity Activity, and IV measurements.
- Assisted PhD scholars in setting up automated measurement systems & maintaining lab instruments.
- Developed/assisted with **automated experimental measurement systems** and ESP32-based instrumentation for laboratory applications.

MAJOR ACADEMIC PROJECTS

- DEVELOPMENT OF A STANDALONE SYSTEM FOR COMBINED VAN DER PAUW RESISTIVITY & HALL EFFECT MEASUREMENTS** 

Final Year Major Project (Integrated M.Sc.) | Prof. Vipul Kheraj | Physics Department, SVNIT Surat

Oct 2023 – Jan 2025

- Designed & developed a **fully automated system** for Van der Pauw resistivity & Hall effect measurements, integrating linear actuators, permanent magnets, & embedded electronics.
- Created a user-friendly interface using a **Nextion touchscreen display** for easy measurement selection and initiation.
- Engineered precision control of magnet positions via automated movement sequences, eliminating the need for manual configuration.
- Developed embedded software to compute electrical parameters & **export results in .txt & .csv** formats.
- Built a **SoftAP-based Wi-Fi** interface to host a local webpage with QR code access, enabling users to download data directly to their devices.
- Achieved high measurement reliability with a low coefficient of variation (CV = 0.19%) and supported by strong normality (Shapiro-Wilk p = 0.973). This ensures the system's robustness and suitability for both academic and industrial material characterization labs.

- EXPERIMENTAL SETUP FOR IN-SITU OPTICAL MONITORING OF NUCLEATION USING LASER TRANSMISSION**

Mini-Project – In collaboration with research activities led by Prof. Vipul Kheraj | Physics Department, SVNIT Surat

Sep 2024

- Conceptualized and developed a real-time optical monitoring system to investigate nucleation kinetics in saturated NaCl-ethanol systems, enabling continuous, **non-invasive tracking and quantitative analysis of phase transitions**.
- Engineered a dual-photodiode laser setup for transmitted-intensity measurement and laser-stability referencing.
- Designed a **custom LabVIEW-based virtual instrument (VI)** to interface **NI VirtualBench modules**, providing synchronized data acquisition, real-time signal visualization, & automated logging.
- Achieved **high temporal resolution optical monitoring** using a precision voltage-divider photodetector circuit, eliminating the need for external amplification.

INDEPENDENT TECHNICAL PROJECTS

- DEVELOPMENT OF A LAB-SCALE IoT SENSOR NETWORK FOR ENVIRONMENTAL & ENERGY MONITORING**

Embedded & IoT Systems Development | Dr. Anupama Kowli, Associate Professor | Electrical Engineering Dept., IIT Bombay

Oct – Nov 2025

- Developed an **IoT device ecosystem** (“Sense Series”) with MQTT-enabled data pipelines on ESP32/ESP8266 and Raspberry Pi platforms.
- Designed modular **IoT sensing devices**—**THSense** (temperature-humidity sensing), **AQSense** (air quality sensing), **ENSENSE** (energy monitoring), and **ACSense** (energy monitoring with control for air conditioners)—using SHT40, BME280, SCD40, MPM10 laser particulate, and PZEM modules.
- Built a Linux server stack with **Apache → MySQL → InfluxDB** migration and **Grafana** visualization.
- Created a distributed **MQTT network** using **Raspberry Pi 5 (Home Assistant)** as a broker and provisioning hub.
- Implemented secure **OTA firmware updates**, enabling remote device upgrades and long-term maintenance without physical access.
- Real-time sensor **telemetry**, database integration, and dashboard automation for lab-wide monitoring.
- Ensured **24×7** uptime, database integrity, and automated device health monitoring across deployed systems.

- AUTONOMOUS LANE-FOLLOWING RC CAR USING COMPUTER VISION & ESP32 MICROCONTROLLER**

Mini-Project – Experimental Embedded Systems & Computer Vision | SVNIT Surat

Dec 2023

- Engineered a miniature autonomous vehicle that performs **real-time lane detection & path correction** using a vision-based guidance system. Leveraged a smartphone as a mobile visual sensor to capture the road view, transmitting **live video to a host system** for processing via **OpenCV**.
- Designed and implemented image processing algorithms using **edge detection** and **perspective transformation** to extract lane boundaries and compute steering angles for autonomous navigation.
- Developed a **custom ESP32 firmware** to receive directional commands over serial & control motor actuation with precise timing. The project serves as a proof-of-concept for cost-effective **Advanced Driver Assistance Systems (ADAS)**, showcasing a seamless integration of edge computing, embedded control, & intelligent mobility.

ACHIEVEMENTS

- Awarded Best Project Recognition** at the **Research & Innovation Conclave 2025**
Organized by the Research & Innovation Affairs Council in collaboration with the Research & Consultancy Section, SVNIT Surat Apr 2025
 - Project recognized & presented during the panel discussion featuring top industry leaders such as **Dr. Sukhdeep Singh** (Samsung R&D), **Mr. Tanmay Bunkar** (BotLab Dynamics), & **Mr. Abhay Salot** (J.P. Morgan Chase), highlighting cutting-edge innovations in research & technology.
- Received a grant of **₹1.3 Lakh (~\$1,550 / ~€1,450)** from the Association for Harnessing Innovation & Entrepreneurship (ASHINE), SVNIT Surat Oct 2023
Under the Student Startup and Innovation Policy (SSIP 2.0) as a part of the IGNITION 3.0 Program launched by the Government of Gujarat
 - Developed a Standalone System for Combined Van der Pauw Resistivity & Hall Effect Measurements.
 - Patent Application Filed** – Application No.: 202521099022, dated October 14, 2025.

RESEARCH & WORKING STYLE

- Independent experimental design** and end-to-end system integration
- Cross-disciplinary** collaboration across physics and electronic systems
- Methodical** experimental execution ensuring reproducible results
- Long-duration** system deployment, monitoring, and reliability testing

AVAILABILITY & MOBILITY

- Open to relocation and international research environments

TECHNICAL PROFICIENCIES

- Embedded Systems & IoT Architecture**
 - Microcontrollers: **ESP32**, ESP8266 (NodeMCU), M5Stack, Arduino
 - Communication Protocols: **MQTT** (Broker & Client), **HTTP/REST**, **LoRa**, **TCP/IP**, **Telnet**
 - OTA** Firmware Updates; **Wi-Fi Provisioning**; **NVS**-based State Persistence
 - Multi-node Sensor Networks & Device Telemetry Pipelines; Home Assistant Integration; **Raspberry Pi** as Automation Hub
- Server, Backend & Data Infrastructure**
 - Linux Server Administration (Ubuntu)
 - Web & Databases: **Apache**, **MySQL**, **InfluxDB**
 - Time-Series** Data Optimization & **Real-Time** Ingestion
 - Visualization & Monitoring: **Grafana Dashboards**, Alerts, API Integration
 - Device Logging & **Long-Term** Data Reliability
- Electronics & Embedded Hardware**
 - High-Resolution ADCs (**24-bit**, **32-bit**)
 - Nextion **Touch Display** Integration
 - Sensor Interfacing: **SHT35&45**, **BME280**, **DHT22**, **SCD40&41**, **SPS30**, **PZEM** (AC Modules), M5Stack sensor modules (TVOC/CO₂, AirQ)
 - SMD & Through-Hole Soldering**; **PCB Assembly**
- Measurement & Test Instrumentation**
 - SR830** Lock-in Amplifier | **Keithley SourceMeter** | **NI VirtualBench**
- Material Fabrication & Processing**
 - Glove Box; Slot-Die Coater; Spin Coater
 - Probe Sonicator; Vacuum Tweezers
- Characterization Techniques**
 - UV-Visible Spectroscopy**
 - Photoluminescence (PL)**
 - Solar Simulator & **I-V Curve Tracing**
 - X-Ray Diffraction (XRD)**
- Software, Programming & Tools**
 - Programming: **Python** (data acquisition, automation scripts, visualization), **C/C++** (embedded firmware development)
 - Engineering & Scientific Tools: **NI LabVIEW**, **SCAPS**, **OceanView**, **Origin**, **Mathematica**
 - Version Control: **Git**, **GitHub**
 - Development Environments: **Arduino IDE**, **Nextion Editor**
 - Documentation: **LaTeX**
- Data Handling & Documentation**
 - CSV** Data Automation & Experimental Log Structuring
 - MS Excel** (Data Processing & Analysis)
 - MS Word & PowerPoint** (Professional Technical Documentation)

WORKSHOPS & SHORT-TERM COURSES

- CENSE WINTER SCHOOL ON SEMICONDUCTOR TECHNOLOGY** 9 – 20 Dec 2024
Conducted by the Centre for Nanoscience & Engineering (CeNSE), IISc Bangalore – Online
- GIAN COURSE – ADVANCES IN PHYSICS & TECHNOLOGY OF LASER-ASSISTED MATERIALS PROCESSING** 8 – 12 Jan 2024
Organized under the Ministry of Education's GIAN Scheme
- CONTROL OF POWER ELECTRONIC CONVERTERS IN RENEWABLE & TRANSPORTATION APPLICATIONS** 28 Sep – 2 Oct 2023
Short-Term Course (STC) sponsored by NaMPET Phase-III, Organized by Dept. of Electrical Engineering, SVNIT Surat

RELEVANT COURSES

- 1. IIT BOMBAY:**
VLSI Technologies (EE669) by **Prof. Anil Kottantharayil** | Solid State Devices (EE733) by **Prof. Udayan Ganguly** | Introduction to Micro-Electro-Mechanical Systems (EE701) by **Prof. Ashwin Seshia** | Sensors in Instruments (EE617) by **Prof. Siddhartha Duttagupta** | Reliability and Failure Analysis of Electronic Devices (EE765) by **Prof. Narendra Shiradkar**
- 2. SVNIT SURAT:**
Digital Electronics (PH306) & Microprocessors (PH432) by **Prof. Vipul Kheraj** | Thin Films & Vacuum Technology (PH424) by **Prof. Dimple V. Shah**

LANGUAGES

- ENGLISH:** FLUENT | **MALAYALAM:** NATIVE | **HINDI:** PROFICIENT

REFERENCES

- 1. PROF. VIPUL KHERAJ**
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